

**Smart Learn: An Adaptive Mobile Platform for Neurodevelopmental
Learning Disorders**

R26-IT-114

Project Proposal Report

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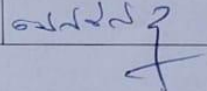
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Dyslexia is a common neurodevelopmental learning disorder affecting children's ability to read accurately and fluently, despite normal intelligence. Children with dyslexia often face persistent difficulties in phonological awareness, letter–sound correspondence, word decoding, and reading comprehension, which can lead to academic struggles, reduced self-confidence, and emotional distress. In Sri Lanka, early intervention for Sinhala-medium children is often delayed due to limited access to trained specialists, high assessment costs, and the absence of culturally and linguistically appropriate tools. Additionally, therapists frequently face challenges in guiding children's reading practice at home, limiting consistent support outside clinical settings.

To address these gaps, this research proposes a web-based mobile application for Sinhala-medium children aged 6–8 years, designed for use under therapist supervision or parental guidance. The application introduces beginner-level literacy games and employs an AI-based voice recognition module, allowing children to read words aloud. Spoken responses are converted to text in real-time, compared with the displayed word, and evaluated for accuracy and pronunciation time. Based on performance, the system dynamically adjusts the difficulty of subsequent tasks, ensuring personalized, adaptive learning. The platform also provides dashboards for therapists and parents to monitor progress, identify common error patterns, and guide practice at home.

The application is developed using React for the interface, Node.js with Express for backend services, and MongoDB/Firebase for secure data storage. Expected outcomes include measurable improvements in reading fluency, decoding skills, and pronunciation, as well as increased learner engagement and stronger parental and therapist involvement. Socially, the system promotes equitable access to early literacy intervention for Sinhala-speaking children, while commercially, it offers a scalable, low-cost, culturally relevant tool for educational and therapeutic institutions.

Keywords: Dyslexia, Adaptive Learning, Voice-Based Reading Intervention, Sinhala-Medium, Mobile Health Technology

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me throughout the completion of my fourth-year research project.

First and foremost, I would like to thank our Research Project Supervisor, Mrs. Bagyani Chathurika, for her continuous guidance, encouragement, and valuable support throughout the entire research process. Her advice, expertise, and constructive feedback played a significant role in the successful completion of this project.

I would also like to extend my sincere appreciation to our Co-Supervisor, Dr. Kaushalya Dissanayake, for her valuable insights, suggestions, and guidance provided during the planning and development stages of this research.

I would like to express my special thanks to Mrs. Kanishaka, therapist, for sharing her professional knowledge and valuable insights regarding dyslexia and learning intervention strategies. Her guidance greatly helped us understand the practical challenges faced by children with reading difficulties.

My sincere appreciation also goes to the healthcare professionals, educational therapists, and pediatric experts who kindly shared their knowledge and experience, which greatly contributed to improving the quality of this research.

I would also like to thank my group members for their teamwork, cooperation, and dedication throughout this project. Their collaboration and support were essential in completing the research successfully.

Finally, I would like to express my heartfelt gratitude to my family and friends for their encouragement, patience, and continuous support during the entire research journey.

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LIST OF ABBREVIATIONS

Abbreviation	Word/Description
NLDs	Neurodevelopmental learning disorders
SDGs	Sustainable Development Goals
AR	Augmented Reality
DSR	Design Science Research
UI	User Interface
VS Code	Visual Studio Code
IDE	integrated development environment
IP	Intellectual Property

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1.INTRODUCTION

1.1 Background and Context

Dyslexia is one of the most common neurodevelopmental learning disorders, affecting millions of school-aged children worldwide. It primarily impairs reading accuracy, phonological awareness, word decoding, spelling, and reading fluency, even in children with normal intelligence who receive standard classroom instruction. Children with dyslexia often struggle with letter–sound matching, blending sounds into words, recognizing familiar words, and comprehending text. Because reading underpins most academic learning, these difficulties can significantly hinder a child’s educational progress, cognitive development, and self-confidence [1].

Early detection and intervention are critical, particularly for children aged 6–8 years, who are in a crucial stage of developing foundational literacy skills. Delays in intervention during this period can lead to long-term academic underachievement, reduced motivation to learn, poor reading fluency, and limited educational opportunities [2]. In many low-resource and underserved regions, including rural areas of Sri Lanka, children often remain undiagnosed due to the high cost of professional assessments, limited availability of trained specialists, and lack of accessible literacy intervention tools designed for Sinhala-medium learners.

Existing support systems include therapist-guided literacy training, phonics-based remedial programs, and digital reading applications. However, many of these approaches have several limitations. They often require continuous supervision by specialists, which can be difficult for families with financial constraints. Additionally, therapists frequently face challenges in guiding children’s reading practice at home, limiting the effectiveness of intervention outside clinical sessions. Although some digital tools provide reading exercises, most lack adaptive learning capabilities, pronunciation evaluation, and real-time performance analysis tailored to dyslexia-specific difficulties.

To address these limitations, this research proposes a web-based mobile learning system for Sinhala-medium children aged 6–8 years, designed to support reading improvement through engaging and adaptive literacy activities. The system is intended to be used under the guidance of therapists, while also allowing children from financially constrained families to use the platform under parental supervision at home.

Key Components of the Proposed System

AI-Based Voice-Interactive Reading Assessment (Novelty):

The novelty of this research lies in the development of an AI-based Sinhala voice-interactive dyslexia learning system that evaluates children’s reading ability using speech recognition and pronunciation time analysis. The system displays Sinhala words and captures the child’s spoken pronunciation through voice input. The recorded speech is converted into text using speech-to-text technology and compared with the expected word to determine pronunciation accuracy. Additionally, the system measures the time taken by the child to pronounce each word, enabling the detection of reading delays commonly associated with dyslexia. This

approach provides a real-time and automated evaluation of pronunciation accuracy and reading fluency, which is currently lacking in most existing dyslexia support applications designed for Sinhala-speaking children.

Adaptive Game-Based Learning:

The system introduces interactive literacy games starting from beginner-level tasks to assess the child's initial reading ability. Based on the child's performance, including pronunciation accuracy and response time, the application dynamically adjusts the difficulty level of subsequent tasks to ensure personalized and progressive learning.

Therapist and Parent Monitoring:

The platform provides progress dashboards for therapists and parents, allowing them to monitor performance trends, identify reading difficulties, and guide the child's learning process both in therapy sessions and at home.

By integrating artificial intelligence, speech recognition, and adaptive learning techniques, the proposed system aims to create a scalable and accessible solution for improving reading skills among Sinhala-speaking children with dyslexia

This solution supports the United Nations Sustainable Development Goals (SDGs), particularly SDG 4 (Quality Education) by promoting inclusive and equitable literacy support, SDG 3 (Good Health and Well-Being) by enhancing cognitive and emotional development, and SDG 10 (Reduced Inequalities) by increasing access to affordable intervention tools for underserved communities [10].

The proposed system aligns with the FoC Research Cluster: Technology Integration & Management, as it combines rule-based artificial intelligence techniques with educational technology (EdTech) and digital health concepts to create a scalable, intelligent, and accessible intervention platform for children with dyslexia.

1.2 Literature Review

In recent years, mobile technologies have emerged as effective tools for supporting dyslexia detection and intervention among children. Researchers have increasingly explored mobile applications, game-based learning environments, and digital screening tools to make dyslexia assessment and literacy training more accessible, interactive, and engaging for young learners. Existing research on dyslexia-related mobile applications can generally be categorized into two major areas: dyslexia screening tools and dyslexia intervention systems.

Several studies have focused on the development of mobile applications designed for dyslexia screening and early identification. In Sri Lanka, a mobile application named "The CURE" was developed to assist individuals with dyslexia and dysgraphia. The study involved approximately 100 participants, including patients and healthcare professionals, and the results showed that the application successfully supported improvements in reading and writing skills while also providing a useful platform for early assessment of learning difficulties [3]. Similarly, a puzzle-based mobile application called "DysPuzzle" was developed in Spain to identify potential dyslexic characteristics among pre-reading children aged 3–6 years. Usability

evaluations demonstrated that children easily understood the gameplay and remained highly engaged with the activities, indicating that game-based screening can effectively support early dyslexia detection [4].

In Italy, researchers developed an Android-based screening application to identify reading disabilities among elementary school students. The application was evaluated with 43 participants, and the results indicated that the system could successfully identify students with reading difficulties while maintaining a user-friendly and engaging interface [5]. Likewise, a mobile screening application developed in Brazil aimed to detect dyslexia by conducting reading recognition tests and analyzing the impact of environmental noise during pronunciation. Experimental evaluations comparing dyslexic and non-dyslexic participants demonstrated that the system could successfully distinguish different reading performance patterns [6].

Researchers have also explored cross-platform digital screening tools. For example, the “DycScreen” system developed in Malaysia enabled dyslexia screening across multiple platforms, including mobile devices and desktop computers. Evaluations involving dyslexic and non-dyslexic students indicated that digital screening tools could effectively identify potential dyslexia indicators while maintaining high levels of student engagement [7]. Similarly, the “Dytective” linguistic web-based game developed in Mexico was tested with 243 participants, including individuals diagnosed with dyslexia. The findings demonstrated that game-based linguistic assessments could accurately detect dyslexia-related difficulties without requiring specialized diagnostic equipment [8].

In addition to screening systems, many researchers have proposed mobile applications aimed at dyslexia intervention and literacy improvement. One notable example is the “Dyseggia” mobile game developed in Spain, which provides word-based exercises to help dyslexic children practice reading and spelling skills. Evaluations with dyslexic children showed that the application was more engaging and motivating than traditional paper-based exercises [9]. Another intervention system, “Mr Read,” developed in Malaysia, was designed to improve reading skills among dyslexic children through multimedia learning activities. Usability testing involving teachers and students indicated that the application was easy to use and effective in supporting reading practice [10].

Further improvements were introduced in “Mr Read V2.0,” an enhanced version of the original application that incorporated quizzes and interactive exercises to strengthen reading comprehension and word recognition skills. Evaluations involving teachers, parents, and students reported noticeable improvements in children's reading abilities and overall user satisfaction [11]. Similarly, the “Prosodiya” mobile serious game focused on improving syllable stress awareness related to reading and spelling skills. The system was evaluated with 137 children diagnosed with dyslexia, and the results showed measurable improvements in literacy performance along with strong user engagement [12].

Another intervention application called “EasyLexia,” developed in Greece, was designed to support students with learning disabilities through interactive exercises aimed at improving phonological decoding and word recognition skills. Evaluation results involving students and

parents indicated increased learner engagement and improvements in reading practice [13]. Additionally, a mobile game-based intervention system developed in Brazil targeted visual, auditory, and working memory skills associated with dyslexia. Usability tests conducted with children and speech therapists showed that the system successfully motivated learners while supporting consistent literacy training [14].

Despite these advancements, several limitations remain in existing dyslexia support applications. Many systems focus primarily on screening and detection rather than providing continuous intervention and adaptive learning support. Additionally, most applications rely heavily on visual interaction methods, such as multiple-choice questions or letter recognition tasks, without incorporating speech-based pronunciation evaluation. Another significant limitation is that the majority of existing applications are developed primarily for English or other widely used languages, with limited support for low-resource languages such as Sinhala. Furthermore, current tools provide limited capabilities for real-time reading performance analysis, response time measurement, and therapist-guided monitoring, which are essential for effective dyslexia intervention.

Therefore, there is a strong need for a personalized, adaptive learning system that integrates voice-based pronunciation analysis, real-time reading evaluation, and game-based literacy training, specifically designed for Sinhala-speaking children aged 6–8 years. Addressing these limitations can significantly improve accessibility to dyslexia intervention tools while supporting therapists and parents in guiding children's literacy development more effectively.

1.3 Research Gap

1. Clearly State What is Missing

Although many digital applications and mobile systems have been developed to support dyslexia screening and literacy development, several important features remain missing in current solutions. Most existing systems mainly focus on basic screening tests or simple visual exercises, such as letter recognition and spelling activities. These systems rarely integrate voice-based pronunciation assessment, adaptive learning mechanisms, or real-time performance analysis. Additionally, many applications lack interactive game-based learning activities that adapt to the learner's performance and difficulty level. Support for low-resource languages such as Sinhala is also very limited, as most available dyslexia tools are designed primarily for English-speaking learners.

2. Why Current Approaches Are Insufficient

Current dyslexia support systems often use static learning content, where the same exercises are delivered to every child regardless of their individual reading abilities or learning challenges. This approach neglects the individual differences among dyslexic learners, such as unique pronunciation errors, reading speed variations, and cognitive processing difficulties. As a result, one-size-fits-all learning designs cannot effectively address each child's specific learning needs.

Furthermore, many existing systems do not provide voice-based instructions or speech-interactive learning environments, which can be particularly helpful for young children with reading difficulties. Another limitation is the lack of parent or guardian monitoring systems, as most applications do not offer dashboards that allow parents or educators to track the child's progress, reading accuracy, or learning improvements over time. Without these monitoring capabilities, it becomes difficult for parents and teachers to support the child's learning process outside the application environment.

3. Justification of Novelty and Creativity

To address these limitations, this research proposes an AI-based Sinhala voice-interactive dyslexia learning system that combines speech recognition, pronunciation accuracy analysis, and adaptive game-based learning. The system will display Sinhala words and capture the child's pronunciation using voice input. The spoken response will be converted into text through speech-to-text technology, and the system will compare it with the expected word to evaluate pronunciation accuracy.

In addition, the system will measure the time taken by the child to pronounce each word, which helps identify reading delays commonly associated with dyslexia. Based on the child's performance and the specific words where the child struggles, the system will identify the learner's difficulty level and automatically adjust the learning activities. The platform will also include interactive game-based exercises designed to improve phonological awareness, reading fluency, and memory skills.

Another innovative feature of the proposed system is the inclusion of voice-based instructions, which guide children through the activities and improve accessibility for learners who struggle with reading instructions. Additionally, the system will provide a progress monitoring dashboard for parents or guardians, allowing them to track the child's reading performance, improvement patterns, and learning progress over time.

4. Evidence-Based Argument

Previous research has demonstrated that mobile learning applications and serious games can support dyslexia screening and literacy training. However, most existing systems focus primarily on visual interaction methods or screening assessments rather than comprehensive learning support. Studies also indicate that many dyslexia applications lack adaptive learning features, pronunciation-based evaluation, and personalized learning paths, which are critical for addressing the diverse needs of dyslexic learners. Furthermore, the majority of existing tools are developed for English-language contexts, leaving languages such as Sinhala underrepresented in technological solutions.

Therefore, there is a clear need for a personalized and adaptive dyslexia learning platform that integrates voice-based interaction, pronunciation accuracy evaluation, response time analysis, and intelligent game-based learning activities. Such a system can provide a more effective and accessible intervention approach for Sinhala-speaking children aged 6–8 years, while also supporting parents and educators in monitoring and guiding children's learning progress.

Table 1: System Comparison

System / Study	Screening Detection	Game-Based Learning	Voice Pronunciation Analysis	Pronunciation Time Measurement	Adaptive Learning / Level Detection	Parent Progress Dashboard	Sinhala Language Support
The CURE (2018)	✓	✗	✗	✗	✗	✗	Limited
DysPuzzle (2015)	✓	✓	✗	✗	✗	✗	✗
Dytective (2016)	✓	✓	✗	✗	✗	✗	✗
Dysegxia (2013)	✗	✓	✗	✗	✗	✗	✗
Mr Read (2016)	✗	✓	✗	✗	✗	Limited	✗
EasyLexia (2017)	✗	✓	✗	✗	✗	✗	✗
Proposed System	✓	✓	✓	✓	✓	✓	✓

2. OBJECTIVE

To design and develop an AI-based Sinhala voice-interactive dyslexia learning system within the research project period that enables Sinhala-speaking children aged 6–8 years to practice reading through voice-based activities, evaluates pronunciation accuracy using speech-to-text technology, measures the time taken to pronounce words, and provides adaptive game-based learning tasks to support and improve their reading development.

2.2 Specific Objectives

1.To develop a voice-based pronunciation assessment module that captures children's spoken responses and converts them into text using speech-to-text technology to evaluate pronunciation accuracy of Sinhala words.

2.To measure and analyze pronunciation response time in order to identify reading delays and fluency difficulties commonly associated with dyslexia.

3.To design and implement adaptive game-based learning activities that improve phonological awareness, memory skills, and reading ability for children with dyslexia.

4.To develop an intelligent learner level detection mechanism that identifies the child's difficulty level based on error patterns and adjusts the learning activities accordingly.

5.To integrate voice-based instructions that guide children through learning activities and improve usability for early learners who struggle with reading instructions.

6. To implement a parent or guardian progress monitoring dashboard that provides insights into the child's reading performance, learning progress, and areas requiring improvement.

3. METHODOLOGY

Research Approach

This research follows a Design Science Research (DSR) approach, which focuses on designing, developing, and evaluating a technological artifact to solve a specific real-world problem. In this study, the artifact is an AI-based Sinhala voice-interactive dyslexia learning system designed to support reading development among children aged 6–8 years.

The research process will involve several stages: problem identification, system design, development, implementation, and evaluation. Initially, the reading challenges faced by dyslexic children will be analyzed through literature review and consultation with educators. Based on these findings, the system will be designed and implemented with features such as speech-based pronunciation assessment, adaptive game-based learning activities, and learner performance monitoring. Finally, the system will be evaluated to measure its effectiveness in improving reading fluency and pronunciation accuracy.

Technical Methods and Algorithms

The proposed system will integrate several technical methods to evaluate reading ability and support dyslexia learning.

Speech-to-Text Recognition:

The system will capture the child's voice when pronouncing a Sinhala word and convert the spoken input into text using a speech recognition model.

Pronunciation Accuracy Analysis:

The recognized text will be compared with the expected word displayed on the screen using string similarity algorithms such as Levenshtein Distance to determine pronunciation correctness.

Pronunciation Time Measurement:

The system will record the time taken by the child to pronounce each word, which will help identify delays in reading fluency.

Adaptive Learning Algorithm:

Based on pronunciation accuracy and response time, the system will identify the child's difficulty level and automatically adjust the complexity of subsequent learning activities.

Game-Based Learning Mechanism:

The platform will include interactive dyslexia-related games designed to improve phonological awareness, memory skills, and word recognition.

Tools and Platforms with Justification

Table 2: Tools and Technologies

Tool / Technology	Purpose	Justification
React.js	Frontend development (Web-based mobile application)	Used to develop a responsive and interactive user interface that can run on both mobile devices and desktops through a web browser.
Node.js (Express.js)	Backend development and API services	Handles application logic, manages communication between the frontend, database, and AI modules, and processes user requests efficiently.
Python (TensorFlow / PyTorch)	AI model training and data analysis	Used to train machine learning models for pronunciation evaluation, adaptive learning, and performance analysis due to strong AI and speech processing libraries.
Speech Recognition API (Google Speech-to-Text / Whisper)	Voice-to-text conversion	Converts children's spoken Sinhala words into text, allowing the system to compare pronunciation accuracy with the expected word.
MongoDB / Firebase	Data storage	Stores student profiles, pronunciation results, response times, and learning progress data securely and efficiently.

For development and collaboration, Git is used for version control, with the project repository hosted on GitHub. The system is developed using Visual Studio Code (VS Code), which serves as the main development environment for coding, debugging, and project management.

Data Collection and Ethical Considerations

Data for this research will be collected from two main sources: expert consultation and system-generated performance data.

Expert knowledge will be gathered from educational psychologists, speech therapists, special education teachers, and literacy specialists. Their insights will help identify the most common dyslexia-related reading challenges and guide the design of assessment tasks and intervention activities.

The system will also collect anonymized performance data generated while children interact with the learning activities. This data includes measures such as phonological awareness accuracy, decoding performance, reading fluency, and comprehension scores.

All data collection procedures will follow appropriate ethical research practices. Participation in the study will require informed consent from parents or guardians, and all collected data will remain confidential. Personal identifiers will not be stored, ensuring that children's privacy is protected throughout the research process.

Validation Strategy and Performance Metrics

Once the system is developed, it will be tested with a small pilot group of approximately 5–6 children. This pilot study aims to evaluate the system's technical accuracy, usability, and educational impact.

Technical Validation

The system's ability to assess reading performance and recommend appropriate activities will be compared with expert observations to ensure the accuracy of scoring algorithms and adaptive recommendations.

Intervention Evaluation

The effectiveness of the intervention will be measured by comparing pre-intervention and post-intervention reading performance. Improvements will be assessed in areas such as:

- Phonological awareness accuracy
- Decoding performance
- Reading fluency (words per minute)
- Reading comprehension scores

Usability Evaluation

Usability will be evaluated through several methods, including observing children's engagement during gameplay, collecting feedback from parents, and measuring usability using the System Usability Scale (SUS).

Performance Targets

The system aims to achieve the following outcomes:

- More than 80% accuracy in classifying dyslexia difficulty levels
- Strong consistency between system assessments and expert evaluations
- Measurable improvement in reading fluency and decoding skills after the intervention period
- High engagement and motivation among children while using the learning activities

4.HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system is a web-based mobile application designed to support children with dyslexia through interactive reading activities and AI-based analysis. The system enables children to practice letter recognition, pronunciation, and reading exercises while collecting performance data to assess reading difficulties.

The architecture consists of several main components:

- Frontend Application (React) – Provides the user interface for children, parents, and therapists. It includes interactive learning games, reading tasks, and dashboards.
- Backend Server – Handles application logic, user authentication, data processing, and communication between the frontend and AI services.
- Database System – Stores user profiles, reading performance data, assessment results, and learning progress records.
- AI/ML Module (Python) – Processes speech input and reading performance indicators to analyze pronunciation accuracy and reading patterns.
- API Layer – Connects the frontend, backend, and AI components, enabling smooth data exchange within the system.

This architecture allows the system to deliver adaptive learning experiences and data-driven insights into children's reading abilities.

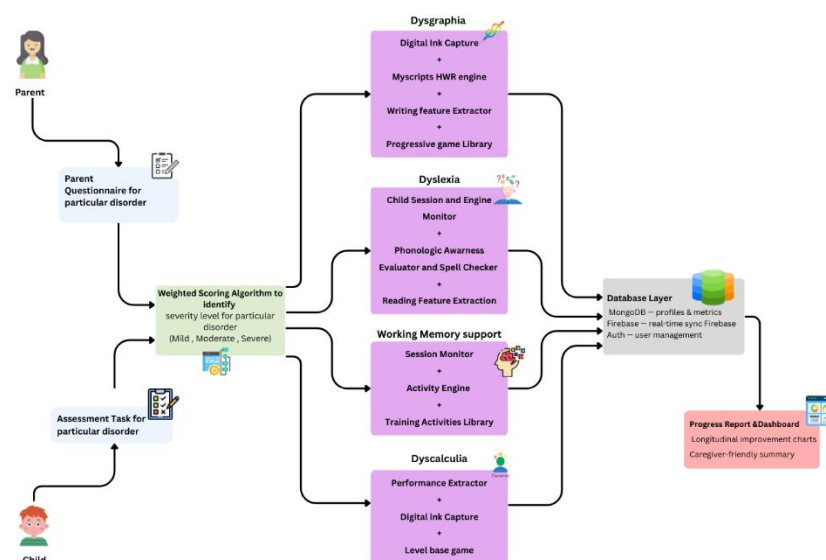


Figure 1: System Diagram

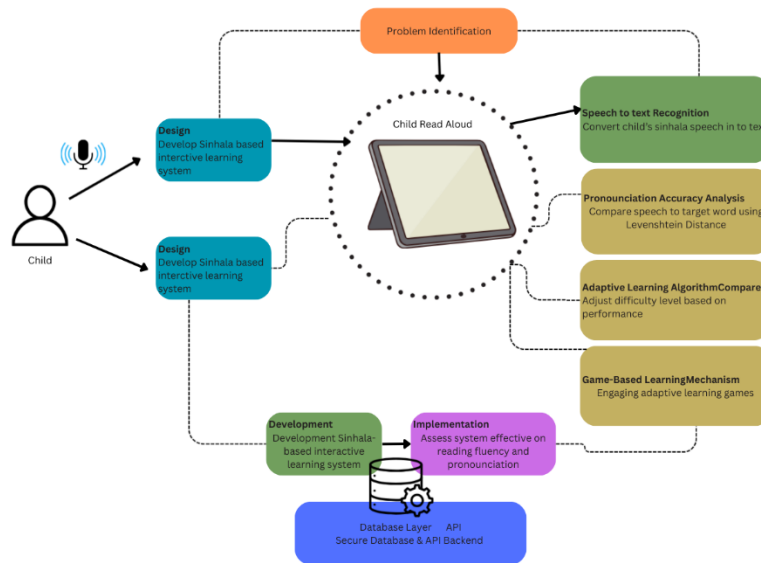


Figure 2:Component diagram

5.USER REQUIREMENTS

Stakeholder identification

The proposed dyslexia intervention system involves several stakeholders who will interact with the system in different ways.

- **Children (Primary Users):**
Children aged 6–8 years who experience reading difficulties related to dyslexia are the primary users of the system. They will use the application to complete reading assessments and participate in interactive learning games designed to improve phonics, decoding, and reading fluency.
- **Teachers Therapists (Secondary Users):** Use the system to guide reading practice and observe student performance
- **Parents / Guardians (Tertiary users):** Monitor the child's progress through the dashboard and support learning at home.

Functional Requirements

The system must support several core functions to assess reading abilities and provide adaptive learning support.

1. The system shall display Sinhala words for children to read aloud.
2. The system shall capture the child's voice input when pronouncing a word.
3. The system shall convert the captured speech into text using speech recognition technology.

4. The system shall compare the recognized text with the expected word to determine pronunciation accuracy.
5. The system shall measure the time taken by the child to pronounce each word.
6. The system shall identify reading difficulties based on pronunciation errors and response time.
7. The system shall provide adaptive game-based learning activities based on the child's performance.
8. The system shall provide voice instructions to guide children through activities.
9. The system shall store learning progress and pronunciation results in a database.
10. The system shall provide a progress monitoring dashboard for parents or guardians.
11. The system shall allow teachers or therapists to review student performance data.
12. The system shall recommend suitable reading games and learning activities based on the child's identified weaknesses and performance results.
13. The system shall provide interactive game-based reading activities to help children practice phonics, word recognition, and reading comprehension skills.

Non-Functional Requirements

Non-functional requirements describe how well the system performs and the quality attributes it must maintain.

1. Usability

The system shall provide a simple, colorful, and child-friendly user interface that enables children aged 6–8 years to interact with the application easily and comfortably.

2. Performance

The system shall provide fast response times during gameplay and pronunciation assessment activities to ensure smooth interaction without noticeable delays.

3. Security

The system shall ensure that user data, including child profiles, voice recordings, and reading performance data, is securely stored and protected from unauthorized access.

4. Reliability

The system shall operate consistently and reliably during learning sessions, minimizing system crashes and preventing loss of user data.

5. Accessibility

The system shall be accessible on commonly used smartphones, tablets, and web browsers, allowing children to use the application easily in both home and educational environments.

6. Scalability

The system architecture shall support future expansion, including the addition of new reading games, learning modules, and support for a larger number of users.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The primary target market for the proposed system is children aged 6–8 years who experience reading difficulties related to dyslexia. Early intervention during primary school years is important because reading skills develop rapidly during this stage.

The main customers who would adopt the system include:

1. Child User Persona

Field	Details
Persona Type	Child Persona (Primary User)
Name	Kavindu
Age	7 years
Role	Grade 2 Student
Location	Sri Lanka
Background	Kavindu is a primary school student who finds it difficult to read Sinhala words and often struggles with letter–sound recognition. He sometimes confuses similar-looking letters and reads more slowly than his classmates.
Needs	Fun and engaging activities to practice reading; audio guidance to hear correct pronunciation; simple learning games to recognize letters and sounds.
Challenges	Difficulty identifying letter sounds; slow reading speed; lack of personalized learning support.
Goals	Improve reading ability; build confidence in learning; learn through interactive games and visual content.

2. Therapist Persona

Field	Details
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Persona Type	Parent Persona
Name	Nadeesha Perera
Age	35 years
Role	Parent of a Grade 2 student
Location	Colombo, Sri Lanka
Background	Nadeesha is concerned that her child is struggling with reading and may have early signs of dyslexia. She wants a reliable tool that can help her support her child's learning at home.
Needs	Early identification of reading difficulties; a structured learning tool to help the child practice reading; progress reports to understand the child's improvement.
Challenges	Limited knowledge about dyslexia; lack of Sinhala-based learning tools; difficulty monitoring the child's reading development.
Goals	Support the child's reading improvement; identify learning difficulties early; provide a comfortable learning environment at home.

3.Parent Persona

Field	Details
Persona Type	Therapist / Educator Persona
Name	Dr. Sanduni Jayasinghe
Age	40 years
Role	Educational Therapist / Special Education Teacher
Location	Sri Lanka
Background	Dr. Sanduni works with children who have learning difficulties such as dyslexia. She provides specialized support to improve reading and phonological skills.
Needs	A digital tool to support therapy sessions; data on children's reading performance; structured activities to strengthen phonological awareness.
Challenges	Limited digital tools for Sinhala dyslexia intervention; difficulty tracking student progress across multiple sessions; need for engaging learning activities for young children.
Goals	Improve children's reading and pronunciation skills; identify dyslexia symptoms early; use technology to support therapy and learning.

The global market for learning disability interventions is substantial, with dyslexia alone affecting one in five children worldwide. In emerging markets like India, approximately 30 million children have special learning needs [16].

Value Proposition

Table 3: Value Proposition

Value Element	Description
Affordable Intervention	Reduces need for expensive ongoing therapy sessions by providing daily practice tool at fraction of cost.
Clear Progress Visibility	Caregiver-friendly reports showing the child's improvement in phonological awareness, word decoding, reading accuracy, and reading fluency.
Inclusive Classroom Support	Enables teachers to support students with handwriting difficulties without specialized training.
Time Savings	Reduces time spent on manual progress documentation and parent updates.
Engaging Game Experience	Writing practice disguised as fun, interactive games with animations and rewards.
Builds Confidence	Positive reinforcement and visible progress reduces reading-related frustration.
Age-Appropriate Design	Bright visuals, friendly characters, and audio prompts designed for 6–8-year-olds.
Peace of Mind	Confidence that child is receiving targeted support for identified difficulties.

Revenue Model

A hybrid revenue model can be adopted:

1. Subscription Model

Monthly or yearly subscription for parents

Tiered plans (Basic, Standard, Premium)

2. Institutional Licensing

Annual licensing for schools and therapy centers.

Multi-user access packages.

3. Freemium Model

Free basic assessment.

Paid access to adaptive intervention modules and progress tracking.

4. B2B Partnerships

Collaboration with pediatric clinics and educational institutions

Cost Estimation

Table 4: Cost Estimation

Cost Category	Estimated Cost (LKR)	Details
Software Development	Rs. 35,000 – 50,000	React.js frontend, Node.js backend, database integration.
Game Design	Rs. 10,000 – 15,000	Child-friendly animations, characters, UI elements for game activities
Expert Consultation	Rs. 10,000 – 15,000	Compensation for occupational therapists and pediatric specialists for validation
Testing and Piloting	Rs. 10,000 – 15,000	Recruitment of participants, incentives, usability testing.
Data Storage	Rs. 10,000 – 15,000	Mongo DB, Firebase
TOTAL INITIAL COSTS	Rs. 65,000 – 100,000	

Pricing Strategy

A penetration pricing strategy can be used initially to attract early users.

- Basic Plan: Low-cost monthly subscription
- Standard Plan: Full intervention access with progress tracking
- Institutional Plan: Discounted bulk pricing

Pricing should remain affordable to ensure accessibility in low-resource environments while maintaining sustainability.

Competitive Advantage

The proposed system offers several advantages compared to existing dyslexia learning applications:

- Support for Sinhala language, which is rarely available in current dyslexia tools.
- Voice-based pronunciation evaluation using speech recognition technology.
- Pronunciation time analysis to detect reading delays.
- Adaptive learning games tailored to the child's performance level.
- Parent monitoring dashboard for tracking learning progress.

These features provide a more personalized and comprehensive dyslexia learning experience, giving the system a strong competitive advantage

Intellectual Property Considerations

The project may involve protecting the software architecture, AI models, and educational content developed for the dyslexia support platform. The system can be protected through copyright registration for the software code, user interface design, and interactive learning game content. If the reading assessment algorithms, speech-based pronunciation analysis, or adaptive learning mechanisms demonstrate sufficient novelty, they may also be considered for patent protection. Additionally, the application name, branding elements, and logo can be

safeguarded through trademark registration to support future commercialization and brand recognition.

7.BUDGET AND JUSTIFICATION

Table 5: Budget and Justification

Cost Category	Estimated Cost (LKR)	JUSTIFICATION
Travel expenses	9,000	Visits to the SLIIT Matara Center to meet the Supervisor and Co-supervisor. Visit to Ayati Center and Other Disability Treatment Centers.
Data collection	15,000	Arrange private appointments to collect data and expert insights regarding dyslexia.
Printing & documentations	1,500	Printing proposals, reports, consent sheets and questionnaires.
Total	25,500	

8.WORK BREAKDOWN STRUCTURE (WBS)

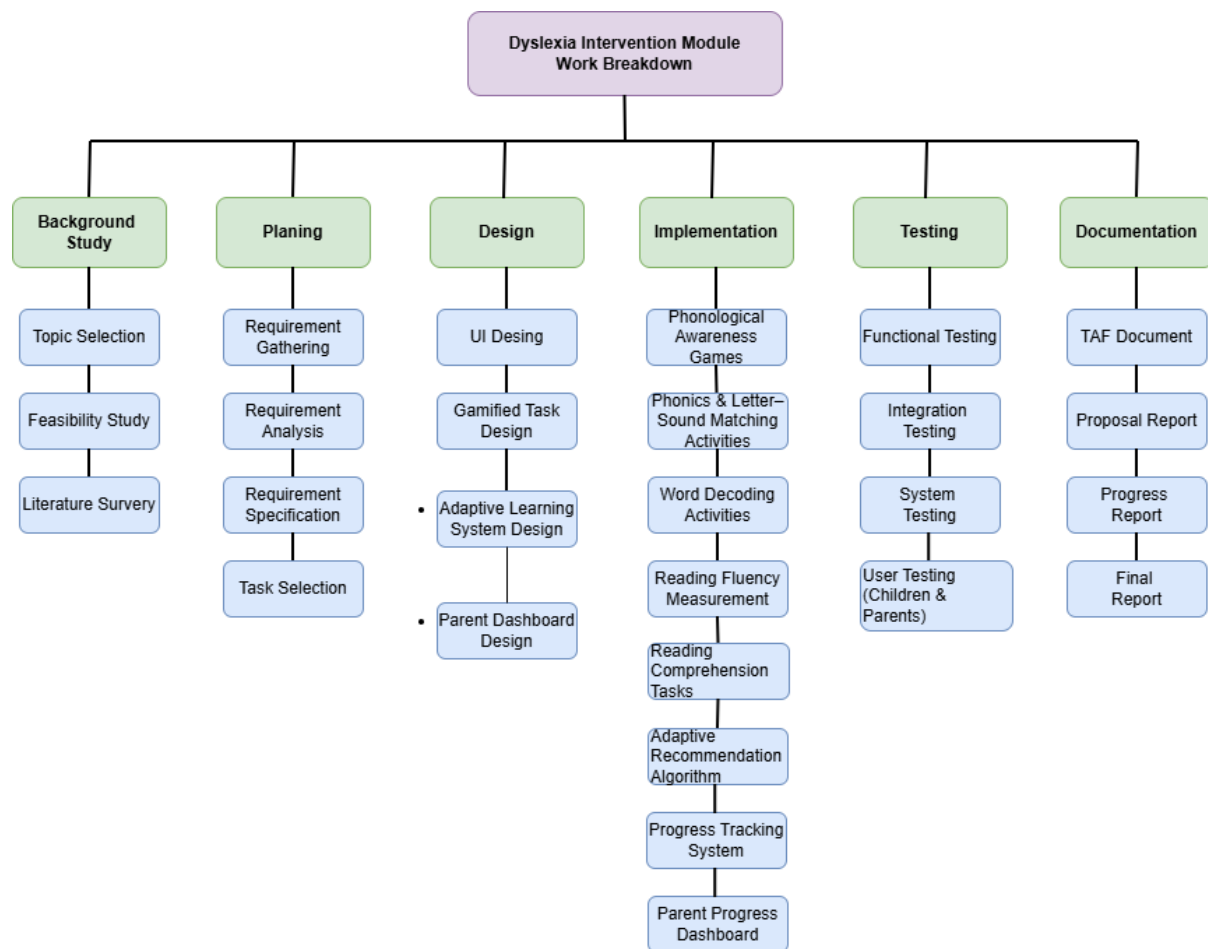
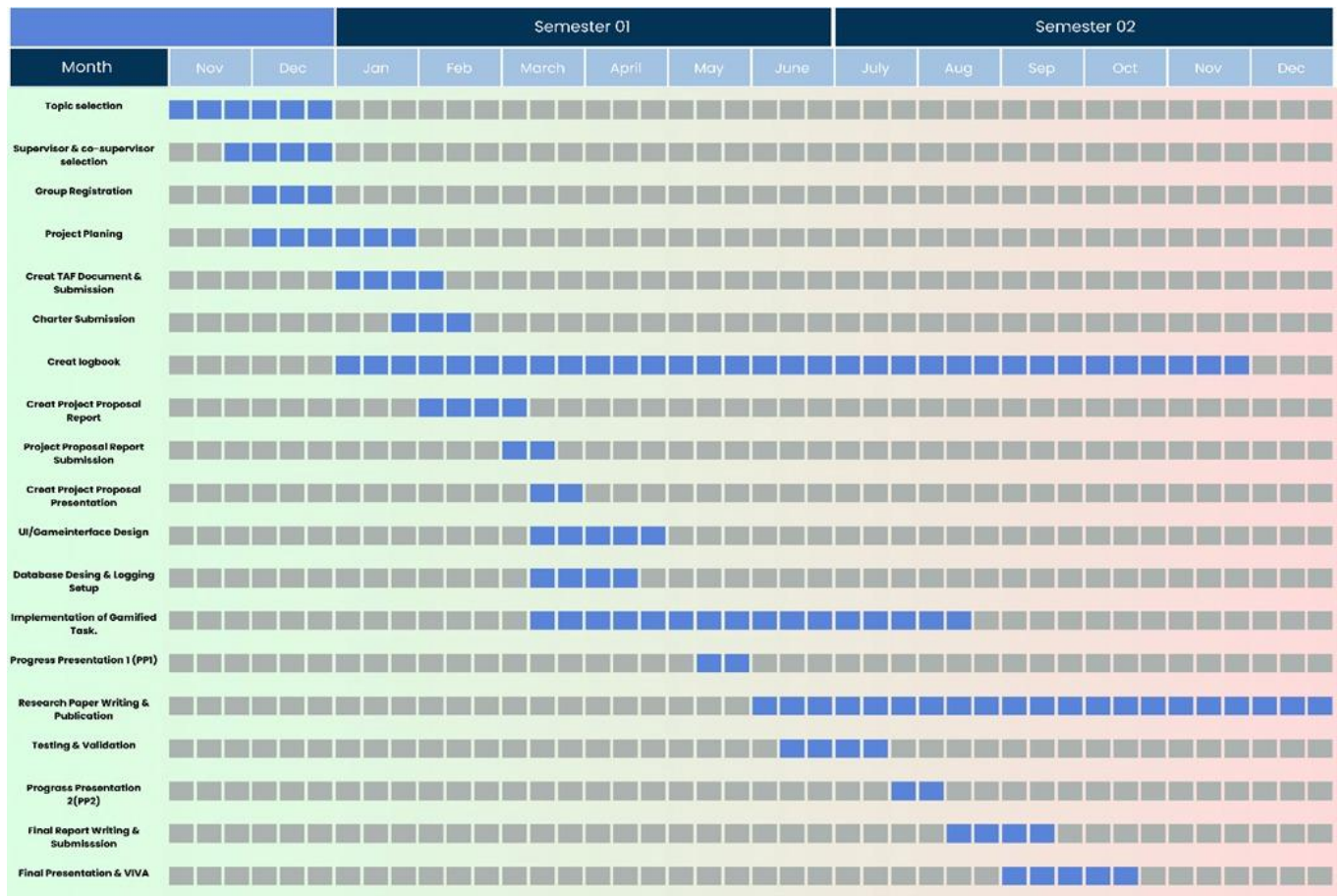


Figure 3:Work Breakdown Chart

9. GANTT CHART

Table 6: Gantt Chart



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11.APPENDICES

Survey instruments

Smart Learn: An Adaptive Mobile Platform for Neurodevelopmental Learning Disorders

Introduction

Smart Learn: An Adaptive Mobile Platform for Neurodevelopmental Learning Disorders is a research and development project conducted by final-year undergraduate students in Information Technology. The project focuses on designing a supportive digital learning application for children with neurodevelopmental learning difficulties such as dyslexia, dyscalculia, dysgraphia, and working memory challenges.

The purpose of this questionnaire is to gather professional insights from therapists, educators, and specialists about current therapy activities, assessment methods, and effective learning strategies for children aged approximately 6–8 years.

The information collected will be used only for academic research and application development purposes. It will help us design a more effective, adaptive, and child-friendly learning application.

All responses will be kept confidential and will only be used to improve the quality, usability, and effectiveness of the proposed mobile platform.

Questions

1. What types of activities or exercises are currently used in therapy for children with dyslexia, dyscalculia, dysgraphia, and working memory difficulties?
2. Are these activities usually paper-based, game-based, digital, or therapist-guided?
3. Which activities do you find most effective for children aged 6–8?
4. How do these activities change as the child improves or struggles?
5. Is there any recognized way to categorize these disorders into levels such as mild, moderate, or severe in clinical practice?
6. Do therapists assess children based on functional ability levels rather than biological markers?
7. How do you typically identify a child’s current level — through standardized tests, observation, questionnaires, or activities?
8. If an app were to adjust its difficulty level based on a child’s performance, what factors should it consider to avoid frustration or overstimulation?

9. What signs indicate a child should move to a higher or lower level?
10. Are there any activities or approaches that should be avoided for children with these disorders?
11. What are the signs that a child is overwhelmed rather than challenged?
12. What kind of progress information is most useful for parents?
13. How do therapists usually communicate improvement to parents?
14. From your professional experience, what is one feature you wish learning apps for neurodevelopmental disorders did better?

Plagiarism Report

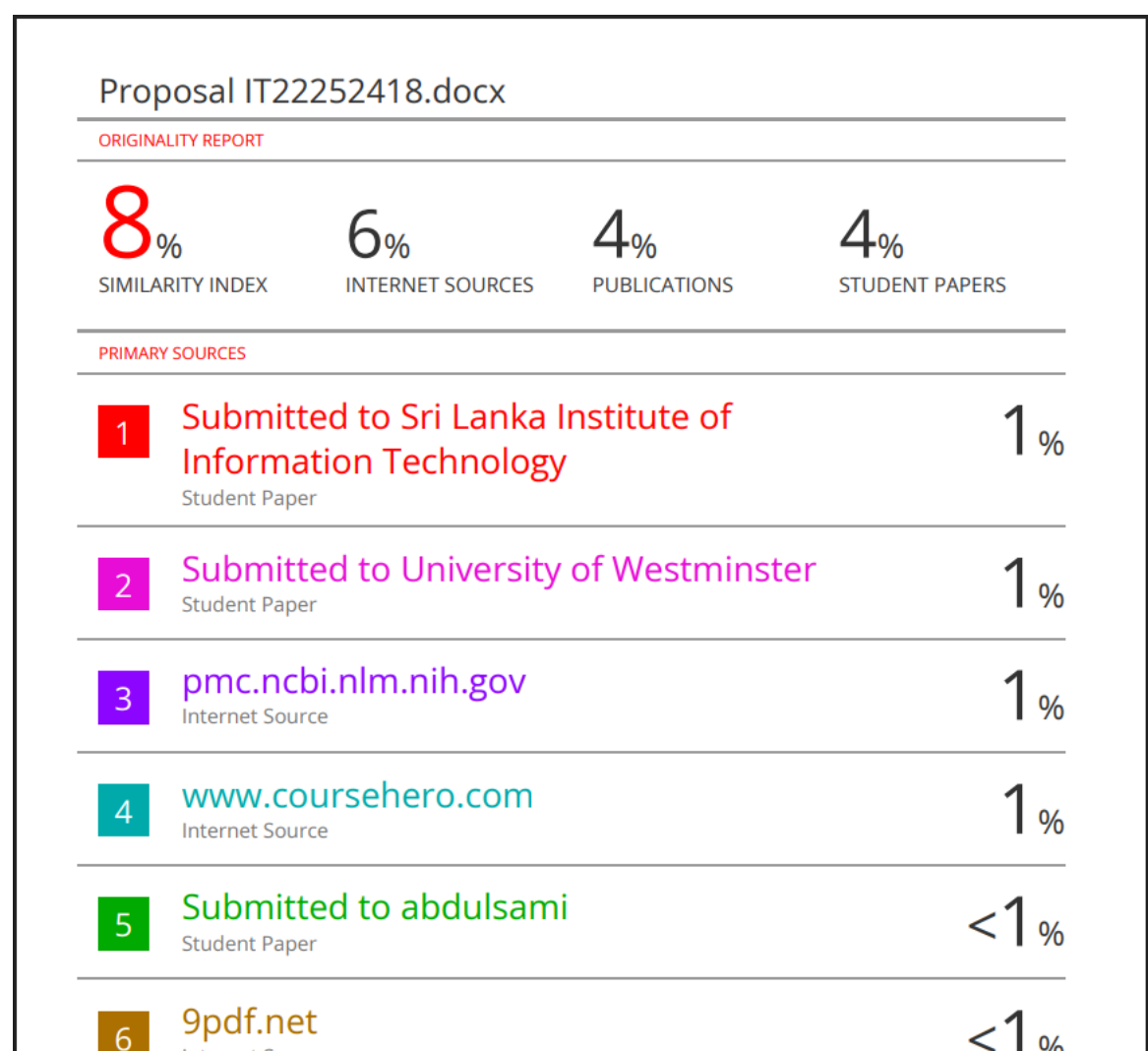


Figure 4:Plagiarism Report

Risk analysis

A risk matrix is used to evaluate potential technical and operational risks that may occur during the development and use of the **adaptive mobile dyslexia intervention system**, including the reading assessment module, adaptive learning engine, and backend services.

Table 7: Risk Analysis

Risk Factor	Impact	Probability	Mitigation Strategy
Inaccurate Reading Skill Assessment (system incorrectly measures phonological awareness or decoding ability)	High	Medium	Use multiple indicators such as decoding accuracy, response time, and comprehension scores. Validate the assessment results with expert feedback and pilot testing with sample users.
Incorrect Dyslexia Severity Classification	High	Low	Implement rule-based scoring thresholds defined with expert consultation. The system will also log assessment results for periodic review and algorithm improvement.
Low Engagement of Children with Learning Activities	Medium	Medium	Use gamification techniques such as rewards, animations, and interactive characters to keep children motivated during reading tasks.
Mobile Device Performance Issues (slow response during games or assessments)	Medium	Low	Optimize the mobile application and use lightweight graphics and efficient data processing to ensure smooth performance on common smartphones.
Data Privacy and Security Risks too easy or too difficult)	Critical	Low	Implement secure authentication, encrypted data transfer (HTTPS), and anonymized storage of children's reading data. Access to progress reports will be restricted to authorized parents or guardians.
Incorrect Adaptive Difficulty Adjustment (system recommends activities that are too easy or too difficult)	Medium	Medium	Continuously analyze user performance and adjust difficulty levels using predefined rules and performance thresholds.

